

Image Hosting Service API

API design reference

OpenAPI 3.0.3 | Version 1.0.0 | Draft

Purpose

A readable companion to docs/openapi.yaml. It summarizes the planned REST endpoints, access rules, request formats, response codes, and reusable schemas for the Image Hosting Service.

BASE URL

http://localhost:8080/api/v1

AUTHENTICATION

Opaque database-backed session cookie named SESSION, managed through Spring Session JDBC.

CORE CAPABILITIES

- Registration, login, logout, and current-user session lookup
- Public and private image upload with a 10 MB limit
- Private object-storage proxy endpoints for original images and thumbnails
- Asynchronous AI tagging with objects, descriptive tags, and colors
- Public browsing, free-text search, pagination, and user galleries

Source of truth: docs/openapi.yaml

API at a glance

The current specification contains 13 core operations across authentication, image management, and public galleries.

13	3	10 MB	50
Operations	Resource groups	Upload limit	Maximum page size

Endpoint index

Method	Path	Access	Purpose
POST	/api/v1/auth/register	Public	Register a new user
POST	/api/v1/auth/login	Public	Log in
POST	/api/v1/auth/logout	Authenticated	Log out
GET	/api/v1/auth/me	Authenticated	Get the current user
GET	/api/v1/images	Public	Browse or search public images
POST	/api/v1/images	Authenticated	Upload an image
GET	/api/v1/images/mine	Authenticated	Get images owned by the current user
GET	/api/v1/images/{imageId}	Public image or owner	Get image metadata
PATCH	/api/v1/images/{imageId}	Owner only	Change image visibility
DELETE	/api/v1/images/{imageId}	Owner only	Delete an image
GET	/api/v1/images/{imageId}/content	Public image or owner	Stream the original image
GET	/api/v1/images/{imageId}/thumbnail	Public image or owner	Stream the 100 x 100 thumbnail
GET	/api/v1/users/{userId}/images	Public	Get a user's public images

Shared conventions

Convention	Value
Base path	/api/v1
Authentication	SESSION cookie; opaque ID stored in PostgreSQL
IDs	UUID strings for users and images
Dates	ISO 8601 date-time values
Pagination	Zero-based page; default size 20; maximum size 50
Public image rules	Public metadata and content are available without a session
Private image rules	Private metadata and content are available only to the owner

Authentication

Session-based account registration and access management.

POST

/api/v1/auth/register

Register a new user

Creates an account, logs the user in automatically, and returns a session cookie.

Property	Value
Access	Public
Operation ID	register
Request	application/json

REQUEST BODY

Content type: application/json

Field	Required	Type	Description
name	Yes	string	min length: 2; max length: 255
email	Yes	string (email)	max length: 255
password	Yes	string (password)	min length: 8; max length: 72

EXAMPLE REQUEST

```
{
  "name": "Shadi",
  "email": "shadi@example.com",
  "password": "StrongPassword123!"
}
```

RESPONSES

Status	Description	Content type
201	User registered and logged in	application/json
400	Invalid request	application/json
409	Email is already registered	application/json

EXAMPLE 201 RESPONSE

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "Shadi",
  "email": "shadi@example.com",
  "createdAt": "2026-07-16T20:30:00Z"
}
```

POST

/api/v1/auth/login

Log in

Property	Value
Access	Public
Operation ID	login
Request	application/json

REQUEST BODY

Content type: application/json

Field	Required	Type	Description
email	Yes	string (email)	
password	Yes	string (password)	

EXAMPLE REQUEST

```
{
  "email": "shadi@example.com",
  "password": "StrongPassword123!"
}
```

RESPONSES

Status	Description	Content type
200	Login successful	application/json
400	Invalid request	application/json
401	Invalid email or password	application/json

EXAMPLE 200 RESPONSE

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "Shadi",
  "email": "shadi@example.com",
  "createdAt": "2026-07-16T20:30:00Z"
}
```

POST

/api/v1/auth/logout

Log out

Property	Value
Access	Authenticated
Operation ID	logout

RESPONSES

Status	Description	Content type
204	Logout successful	—
401	Authentication is required or the session has expired	application/json

	No response body A 204 response confirms success and intentionally contains no JSON body.
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GET	/api/v1/auth/me
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Get the current user

Property	Value
Access	Authenticated
Operation ID	getCurrentUser

RESPONSES

Status	Description	Content type
200	Current authenticated user	application/json
401	Authentication is required or the session has expired	application/json

EXAMPLE 200 RESPONSE

<pre>{ "id": "550e8400-e29b-41d4-a716-446655440000", "name": "Shadi", "email": "shadi@example.com", "createdAt": "2026-07-16T20:30:00Z" }</pre>

Images

Upload, browse, search, manage, and securely stream image content.

GET	/api/v1/images
-----	----------------

Browse or search public images

Without `q`, returns public images from newest to oldest. With `q`, searches AI-generated objects, tags, and colors.

Property	Value
Access	Public
Operation ID	getPublicImages

PARAMETERS

Name	In	Required	Type	Description
q	query	No	string	Free-text search across AI-generated objects, tags, and colors; minLength 1, maxLength 100
page	query	No	integer	Zero-based page number; default 0, min 0
size	query	No	integer	Number of images per page, maximum 50; default 20, min 1, max 50

RESPONSES

Status	Description	Content type
200	Public image page	application/json
400	Invalid request	application/json

EXAMPLE 200 RESPONSE

```
{
  "items": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440000",
      "owner": {
        "id": "550e8400-e29b-41d4-a716-446655440000",
        "name": "Shadi"
      },
      "originalFilename": "beach-sunset.jpg",
      "contentType": "image/jpeg",
      "sizeBytes": 2485190,
      "width": 1920,
      "height": 1080,
      "isPublic": true,
      "aiTags": {
        "objects": [
          "dog",
          "tree"
        ],
        "tags": [
          "outdoor",
          "daytime",
          "playful"
        ],
        "colors": [
          "brown",
          "green",
          "blue"
        ]
      },
      "taggingStatus": "COMPLETED",
      "contentUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/content",
      "thumbnailUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/thumbnail",
      "createdAt": "2026-07-16T20:30:00Z",
      "updatedAt": "2026-07-16T20:30:00Z"
    }
  ],
  "page": 0,
  "size": 20,
  "totalElements": 42,
  "totalPages": 3
}
```

POST

/api/v1/images

Upload an image

Uploads an image of up to 10 MB. Thumbnail generation and AI tagging continue asynchronously after the upload succeeds.

Property	Value
Access	Authenticated
Operation ID	uploadImage
Request	multipart/form-data

REQUEST BODY

Content type: multipart/form-data

Field	Required	Type	Description
file	Yes	string (binary)	JPEG, PNG, or WebP image, maximum 10 MB
isPublic	Yes	boolean	default: False

EXAMPLE FORM FIELDS

```
file: <binary image>
isPublic: false
```

RESPONSES

Status	Description	Content type
201	Image uploaded	application/json
400	Invalid request	application/json
401	Authentication is required or the session has expired	application/json
413	Image exceeds the 10 MB limit	application/json
415	Unsupported format or invalid image content	application/json

EXAMPLE 201 RESPONSE

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "owner": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "name": "Shadi"
  },
  "originalFilename": "beach-sunset.jpg",
  "contentType": "image/jpeg",
  "sizeBytes": 2485190,
  "width": 1920,
  "height": 1080,
  "isPublic": true,
  "aiTags": {
    "objects": [
      "dog",
      "tree"
    ],
    "tags": [
      "outdoor",
      "daytime",
      "playful"
    ],
    "colors": [
      "brown",
      "green",
      "blue"
    ]
  },
  "taggingStatus": "COMPLETED",
  "contentUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/content",
  "thumbnailUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/thumbnail",
  "createdAt": "2026-07-16T20:30:00Z",
  "updatedAt": "2026-07-16T20:30:00Z"
}
```


GET

</api/v1/images/mine>

Get images owned by the current user

Property	Value
Access	Authenticated
Operation ID	getMyImages

PARAMETERS

Name	In	Required	Type	Description
page	query	No	integer	Zero-based page number; default 0, min 0
size	query	No	integer	Number of images per page, maximum 50; default 20, min 1, max 50

RESPONSES

Status	Description	Content type
200	Current user's public and private images	application/json
401	Authentication is required or the session has expired	application/json

EXAMPLE 200 RESPONSE

```
{
  "items": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440000",
      "owner": {
        "id": "550e8400-e29b-41d4-a716-446655440000",
        "name": "Shadi"
      },
      "originalFilename": "beach-sunset.jpg",
      "contentType": "image/jpeg",
      "sizeBytes": 2485190,
      "width": 1920,
      "height": 1080,
      "isPublic": true,
      "aiTags": {
        "objects": [
          "dog",
          "tree"
        ],
        "tags": [
          "outdoor",
          "daytime",
          "playful"
        ],
        "colors": [
          "brown",
          "green",
          "blue"
        ]
      },
      "taggingStatus": "COMPLETED",
      "contentUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/content",
      "thumbnailUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/thumbnail",
      "createdAt": "2026-07-16T20:30:00Z",
      "updatedAt": "2026-07-16T20:30:00Z"
    }
  ],
  "page": 0,
  "size": 20,
  "totalElements": 42,
  "totalPages": 3
}
```

GET

/api/v1/images/{imageId}

Get image metadata

Public images are available to everyone. Private images are available only to their owner.

Property	Value
Access	Public image or owner
Operation ID	getImage

PARAMETERS

Name	In	Required	Type	Description
imageId	path	Yes	string (uuid)	

RESPONSES

Status	Description	Content type
200	Image metadata	application/json
404	Image does not exist or is not accessible	application/json

EXAMPLE 200 RESPONSE

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "owner": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "name": "Shadi"
  },
  "originalFilename": "beach-sunset.jpg",
  "contentType": "image/jpeg",
  "sizeBytes": 2485190,
  "width": 1920,
  "height": 1080,
  "isPublic": true,
  "aiTags": {
    "objects": [
      "dog",
      "tree"
    ],
    "tags": [
      "outdoor",
      "daytime",
      "playful"
    ],
    "colors": [
      "brown",
      "green",
      "blue"
    ]
  },
  "taggingStatus": "COMPLETED",
  "contentUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/content",
  "thumbnailUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/thumbnail",
  "createdAt": "2026-07-16T20:30:00Z",
  "updatedAt": "2026-07-16T20:30:00Z"
}
```

PATCH

/api/v1/images/{imageId}

Change image visibility

Property	Value
Access	Owner only
Operation ID	updateImageVisibility
Request	application/json

PARAMETERS

Name	In	Required	Type	Description
imageId	path	Yes	string (uuid)	

REQUEST BODY

Content type: application/json

Field	Required	Type	Description
isPublic	Yes	boolean	

EXAMPLE REQUEST

```
{
  "isPublic": true
}
```

RESPONSES

Status	Description	Content type
200	Visibility updated	application/json
400	Invalid request	application/json
401	Authentication is required or the session has expired	application/json
403	The authenticated user does not own this resource	application/json
404	Requested resource was not found	application/json

EXAMPLE 200 RESPONSE

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "owner": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "name": "Shadi"
  },
  "originalFilename": "beach-sunset.jpg",
  "contentType": "image/jpeg",
  "sizeBytes": 2485190,
  "width": 1920,
  "height": 1080,
  "isPublic": true,
  "aiTags": {
    "objects": [
      "dog",
      "tree"
    ],
    "tags": [
      "outdoor",
      "daytime",
      "playful"
    ],
    "colors": [
      "brown",
      "green",
      "blue"
    ]
  },
  "taggingStatus": "COMPLETED",
  "contentUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/content",
  "thumbnailUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/thumbnail",
  "createdAt": "2026-07-16T20:30:00Z",
  "updatedAt": "2026-07-16T20:30:00Z"
}
```

DELETE

`/api/v1/images/{imageId}`

Delete an image

Deletes the database record, original object, and thumbnail object.

Property	Value
Access	Owner only
Operation ID	deleteImage

PARAMETERS

Name	In	Required	Type	Description
imageId	path	Yes	string (uuid)	

RESPONSES

Status	Description	Content type
204	Image deleted	—
401	Authentication is required or the session has expired	application/json
403	The authenticated user does not own this resource	application/json
404	Requested resource was not found	application/json

No response body

A 204 response confirms success and intentionally contains no JSON body.

GET

`/api/v1/images/{imageId}/content`

Stream the original image

Property	Value
Access	Public image or owner
Operation ID	getImageContent

PARAMETERS

Name	In	Required	Type	Description
imageId	path	Yes	string (uuid)	

RESPONSES

Status	Description	Content type
200	Original image content	image/jpeg, image/png, image/webp
404	Image does not exist or is not accessible	application/json

Binary response

The response body contains image bytes. The Content-Type is image/jpeg, image/png, or image/webp.

GET

`/api/v1/images/{imageId}/thumbnail`**Stream the 100 x 100 thumbnail**

Property	Value
Access	Public image or owner
Operation ID	getImageThumbnail

PARAMETERS

Name	In	Required	Type	Description
imageId	path	Yes	string (uuid)	

RESPONSES

Status	Description	Content type
200	Thumbnail content	image/jpeg, image/png, image/webp
404	Image or thumbnail does not exist, or is not accessible	application/json
409	Thumbnail generation is not complete	application/json

Binary response

The response body contains image bytes. The Content-Type is image/jpeg, image/png, or image/webp.

Public Gallery

Public browsing of images belonging to a selected user.

GET**/api/v1/users/{userId}/images**

Get a user's public images

Property	Value
Access	Public
Operation ID	getUserPublicImages

PARAMETERS

Name	In	Required	Type	Description
userId	path	Yes	string (uuid)	
page	query	No	integer	Zero-based page number; default 0, min 0
size	query	No	integer	Number of images per page, maximum 50; default 20, min 1, max 50

RESPONSES

Status	Description	Content type
200	User's public gallery	application/json
404	Requested resource was not found	application/json

EXAMPLE 200 RESPONSE

```

{
  "user": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "name": "Shadi"
  },
  "images": {
    "items": [
      {
        "id": "550e8400-e29b-41d4-a716-446655440000",
        "owner": {
          "id": "550e8400-e29b-41d4-a716-446655440000",
          "name": "Shadi"
        },
        "originalFilename": "beach-sunset.jpg",
        "contentType": "image/jpeg",
        "sizeBytes": 2485190,
        "width": 1920,
        "height": 1080,
        "isPublic": true,
        "aiTags": {
          "objects": [
            "dog",
            "tree"
          ],
          "tags": [
            "outdoor",
            "daytime",
            "playful"
          ],
          "colors": [
            "brown",
            "green",
            "blue"
          ]
        },
        "taggingStatus": "COMPLETED",
        "contentUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/content",
        "thumbnailUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/thumbnail",
        "createdAt": "2026-07-16T20:30:00Z",
        "updatedAt": "2026-07-16T20:30:00Z"
      }
    ],
    "page": 0,
    "size": 20,
    "totalElements": 42,
    "totalPages": 3
  }
}

```


Reusable data schemas

The schemas below are shared across endpoint requests and responses. Required fields are marked explicitly.

RegisterRequest

Field	Required	Type	Notes
name	Yes	string	Min length: 2; Max length: 255
email	Yes	string (email)	Max length: 255
password	Yes	string (password)	Min length: 8; Max length: 72

EXAMPLE

```
{
  "name": "Shadi",
  "email": "shadi@example.com",
  "password": "StrongPassword123!"
}
```

LoginRequest

Field	Required	Type	Notes
email	Yes	string (email)	
password	Yes	string (password)	

EXAMPLE

```
{
  "email": "shadi@example.com",
  "password": "StrongPassword123!"
}
```

AuthenticatedUser

Field	Required	Type	Notes
id	Yes	string (uuid)	
name	Yes	string	
email	Yes	string (email)	
createdAt	Yes	string (date-time)	

EXAMPLE

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "Shadi",
  "email": "shadi@example.com",
  "createdAt": "2026-07-16T20:30:00Z"
}
```

PublicUser

Field	Required	Type	Notes
id	Yes	string (uuid)	
name	Yes	string	

EXAMPLE

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "Shadi"
}
```

UpdateImageVisibilityRequest

Field	Required	Type	Notes
isPublic	Yes	boolean	

EXAMPLE

```
{
  "isPublic": true
}
```

ImageTags

Field	Required	Type	Notes
objects	Yes	array<string>	
tags	Yes	array<string>	
colors	Yes	array<string enum>	Max items: 3

EXAMPLE

```
{
  "objects": [
    "dog",
    "tree"
  ],
  "tags": [
    "outdoor",
    "daytime",
    "playful"
  ],
  "colors": [
    "brown",
    "green",
    "blue"
  ]
}
```

ImageResponse

Field	Required	Type	Notes
id	Yes	string (uuid)	
owner	Yes	PublicUser	
originalFilename	Yes	string	
contentType	Yes	string	
sizeBytes	Yes	integer (int64)	
width	Yes	integer	
height	Yes	integer	
isPublic	Yes	boolean	
aiTags	No	ImageTags	Nullable
taggingStatus	Yes	string enum	Allowed: PENDING, PROCESSING, COMPLETED, FAILED
contentUrl	Yes	string	
thumbnailUrl	No	string	Nullable
createdAt	Yes	string (date-time)	
updatedAt	No	string (date-time)	Nullable

EXAMPLE

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "owner": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "name": "Shadi"
  },
  "originalFilename": "beach-sunset.jpg",
  "contentType": "image/jpeg",
  "sizeBytes": 2485190,
  "width": 1920,
  "height": 1080,
  "isPublic": true,
  "aiTags": {
    "objects": [
      "dog",
      "tree"
    ],
    "tags": [
      "outdoor",
      "daytime",
      "playful"
    ],
    "colors": [
      "brown",
      "green",
      "blue"
    ]
  },
  "taggingStatus": "COMPLETED",
  "contentUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/content",
  "thumbnailUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/thumbnail",
  "createdAt": "2026-07-16T20:30:00Z",
  "updatedAt": "2026-07-16T20:30:00Z"
}
```

ImagePage

Field	Required	Type	Notes
items	Yes	array<ImageResponse>	
page	Yes	integer	Minimum: 0
size	Yes	integer	Minimum: 1; Maximum: 50
totalElements	Yes	integer (int64)	Minimum: 0
totalPages	Yes	integer	Minimum: 0

EXAMPLE

```
{
  "items": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440000",
      "owner": {
        "id": "550e8400-e29b-41d4-a716-446655440000",
        "name": "Shadi"
      },
      "originalFilename": "beach-sunset.jpg",
      "contentType": "image/jpeg",
      "sizeBytes": 2485190,
      "width": 1920,
      "height": 1080,
      "isPublic": true,
      "aiTags": {
        "objects": [
          "dog",
          "tree"
        ],
        "tags": [
          "outdoor",
          "daytime",
          "playful"
        ],
        "colors": [
          "brown",
          "green",
          "blue"
        ]
      },
      "taggingStatus": "COMPLETED",
      "contentUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/content",
      "thumbnailUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/thumbnail",
      "createdAt": "2026-07-16T20:30:00Z",
      "updatedAt": "2026-07-16T20:30:00Z"
    }
  ],
  "page": 0,
  "size": 20,
  "totalElements": 42,
  "totalPages": 3
}
```

PublicUserGalleryResponse

Field	Required	Type	Notes
user	Yes	PublicUser	
images	Yes	ImagePage	

EXAMPLE

```
{
  "user": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "name": "Shadi"
  },
  "images": {
    "items": [
      {
        "id": "550e8400-e29b-41d4-a716-446655440000",
        "owner": {
          "id": "550e8400-e29b-41d4-a716-446655440000",
          "name": "Shadi"
        },
        "originalFilename": "beach-sunset.jpg",
        "contentType": "image/jpeg",
        "sizeBytes": 2485190,
        "width": 1920,
        "height": 1080,
        "isPublic": true,
        "aiTags": {
          "objects": [
            "dog",
            "tree"
          ],
          "tags": [
            "outdoor",
            "daytime",
            "playful"
          ],
          "colors": [
            "brown",
            "green",
            "blue"
          ]
        },
        "taggingStatus": "COMPLETED",
        "contentUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/content",
        "thumbnailUrl": "/api/v1/images/550e8400-e29b-41d4-a716-446655440000/thumbnail",
        "createdAt": "2026-07-16T20:30:00Z",
        "updatedAt": "2026-07-16T20:30:00Z"
      }
    ],
    "page": 0,
    "size": 20,
    "totalElements": 42,
    "totalPages": 3
  }
}
```

ValidationError

Field	Required	Type	Notes
field	Yes	string	
message	Yes	string	

EXAMPLE

```
{
  "field": "email",
  "message": "must be a valid email address"
}
```

ErrorResponse

Field	Required	Type	Notes
timestamp	Yes	string (date-time)	
status	Yes	integer	
error	Yes	string	
message	Yes	string	
path	Yes	string	
validationErrors	No	array<ValidationError>	

EXAMPLE

```
{
  "timestamp": "2026-07-16T20:30:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Request validation failed",
  "path": "/api/v1/auth/register",
  "validationErrors": [
    {
      "field": "email",
      "message": "must be a valid email address"
    }
  ]
}
```

Error handling

Errors use a consistent JSON structure so the frontend can display a clear message and field-level validation feedback.

STANDARD ERROR BODY

```
{
  "timestamp": "2026-07-16T20:30:00Z",
  "status": 400,
  "error": "Bad Request",
  "message": "Request validation failed",
  "path": "/api/v1/auth/register",
  "validationErrors": [
    {
      "field": "email",
      "message": "must be a valid email address"
    }
  ]
}
```

Reusable response	Meaning
BadRequest	Invalid request
Unauthorized	Authentication is required or the session has expired
Forbidden	The authenticated user does not own this resource
NotFound	Requested resource was not found

Implementation notes

- The SESSION cookie should be HttpOnly, Secure in production, and use an appropriate SameSite policy.
- Private images should return a non-revealing not-found response when the requester is not the owner.
- Uploaded image content must be validated independently of the client-provided Content-Type.
- Thumbnail generation and AI tagging run asynchronously; URLs or AI tags may be unavailable while processing is pending.
- The object-storage bucket remains private; the API streams image bytes after checking access rules.

Maintenance

When an endpoint, request field, response body, or access rule changes, update docs/openapi.yaml first and regenerate this reference PDF.